



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.NOS.1

## Divide Mixed Numbers

Lesson 6-10 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can divide mixed numbers by first changing them to improper fractions.

**Language Objective:** I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Divide Mixed Numbers

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal



Tap any word to see what it means and a picture.

1

— Change mixed numbers to improper fractions, then multiply by the reciprocal.

2

A number with a whole number and a fraction, like  $2\frac{1}{3}$ , is a

3

A fraction whose numerator is greater than or equal to its denominator, like  $\frac{7}{3}$ , is an

4

Changing a mixed number into an improper fraction is to

it.

5

Writing a fraction in its smallest equal form is to

it.

6

To divide by a fraction, I multiply by its  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How does dividing mixed numbers help the detective?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Divide Mixed Numbers** — Change mixed numbers to improper fractions, then multiply by the reciprocal.  
*Dividir números mixtos — Cambiar los números mixtos a fracciones impropias y luego multiplicar por el recíproco.*

☐ **Mixed Number** — A whole number plus a fraction, like  $2\frac{1}{3}$ .  
*Número mixto — Un número entero más una fracción, como  $2\frac{1}{3}$ .*

☐ **Improper Fraction** — A fraction where the top is bigger than or equal to the bottom, like  $\frac{7}{4}$ .  
*Fracción impropia — Una fracción donde el número de arriba es mayor o igual al de abajo, como  $\frac{7}{4}$ .*

☐ **Convert** — To change a number to a new form but keep the same value.  
*Convertir — Cambiar un número a otra forma sin cambiar su valor.*

☐ **Simplify** — To make a fraction smaller using the same parts, like  $\frac{2}{4} = \frac{1}{2}$ .  
*Simplificar — Hacer una fracción más pequeña con las mismas partes, como  $\frac{2}{4} = \frac{1}{2}$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The detective can cut \_\_\_\_ pieces because  $5\frac{1}{4} \div \frac{3}{4} = 21/4 \times 4/3 = 84/12$  = \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A mixed number must first become an improper fraction so I can flip and multiply.

**Evidence:** Students explain that the mixed number must be converted to an improper fraction ( $7/2$ ) before applying Keep, Change, Flip.

**Reasoning:** This evidence connects the definition of divide mixed numbers to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The detective can cut \_\_\_\_ pieces because  $5 \frac{1}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = \frac{84}{12} = \underline{\hspace{1cm}}$ .**
- **My answer is \_\_\_\_.**

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Divide Mixed Numbers
2. **Notes 2:** Mixed Number
3. **Notes 3:** Improper Fraction
4. **Notes 4:** Convert
5. **Notes 5:** Simplify
6. **Notes 6:** Reciprocal
7. **Watch for this mistake:** *A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first — for example, treating  $4\frac{1}{2} \div 1\frac{1}{2}$  as  $(4 \div 1) + (1/2 \div 1/2) = 5$ . Always convert each mixed number to one improper fraction first, then divide using Keep, Change, Flip:  $4\frac{1}{2} \div 1\frac{1}{2} = 9/2 \div 3/2 = 9/2 \times 2/3 = 3$ .*
8. **Try It 1:** A.  $2 - 3 \div 1\frac{1}{2} = 3 \div 3/2 = 3 \times 2/3 = 6/3 = 2$  sections.
9. **Try It 2:** A.  $3 - 7\frac{1}{2} = 15/2$  and  $2\frac{1}{2} = 5/2$ . So  $15/2 \div 5/2 = 15/2 \times 2/5 = 30/10 = 3$  doorways.